Energy Transformations

Chapter 6 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **How do energy transformations help people?**

KEY WORDS

transform / convert — To transform energy is to change it _____ one form to another.

energy transformation — Energy transformation is when energy _____ form.

KEY IDEA

KEY IDEA **Transformation = energy CHANGES FORM. Transfer = energy MOVES. BOTH can happen together.**

MODEL IT

A device takes energy in, changes its form, and gives a useful output.



QUICK CHECK

Name one device that transforms energy, and tell what form it changes into.

_____

Trace the Energy

Chapter 6 Guided Notes · Page 2

Name: _____ **Date:** _____

DEVICE SYSTEMS

For each device, write the input, the change in form, and the output.

DEVICE	INPUT	TRANSFORMATION	OUTPUT / CHANGE
flashlight	battery (stored)	electrical → light	
buzzer			
motor			
wind-up toy			

TRANSFER VS TRANSFORMATION VS BOTH

Sort each system. Some are BOTH.

SYSTEM	TRANSFER	TRANSFORMATION	BOTH
falling dominoes			
battery → flashlight → light			
warm water → cooler spoon			

Engineering Design

Chapter 6 Guided Notes · Page 3

Name: _____ **Date:** _____

PROBLEM, CRITERIA, CONSTRAINTS

Problem — something we need to _____

Criteria — the things a good design _____

Constraints — the things that _____ what we can use.

ENGINEERING CYCLE

Engineers design, build, test, and improve. Test results — even failures — are useful evidence.

PROBLEM

↓

DESIGN

↓

TEST

↓

EVIDENCE

↓

IMPROVE

MY BLUEPRINT

Sketch your Tabletop Alert / Signal Device. Label the energy pathway.

draw and label your device blueprint

Test → Change → Retest

Chapter 6 Guided Notes · Page 4

Name: _____ **Date:** _____

TEST 1 VS TEST 2

TEST 1 — WHAT HAPPENED?	WHAT I CHANGED (AND WHY)	TEST 2 — WHAT HAPPENED?

DRAW YOUR IMPROVED DESIGN

draw the change you made to your device

CHECK YOUR THINKING

Your first design did not work well. Explain why that is useful evidence, not a failure.

CONNECT *Chapter 1 → 6: energy causes change; devices transform energy to solve problems people have.*